

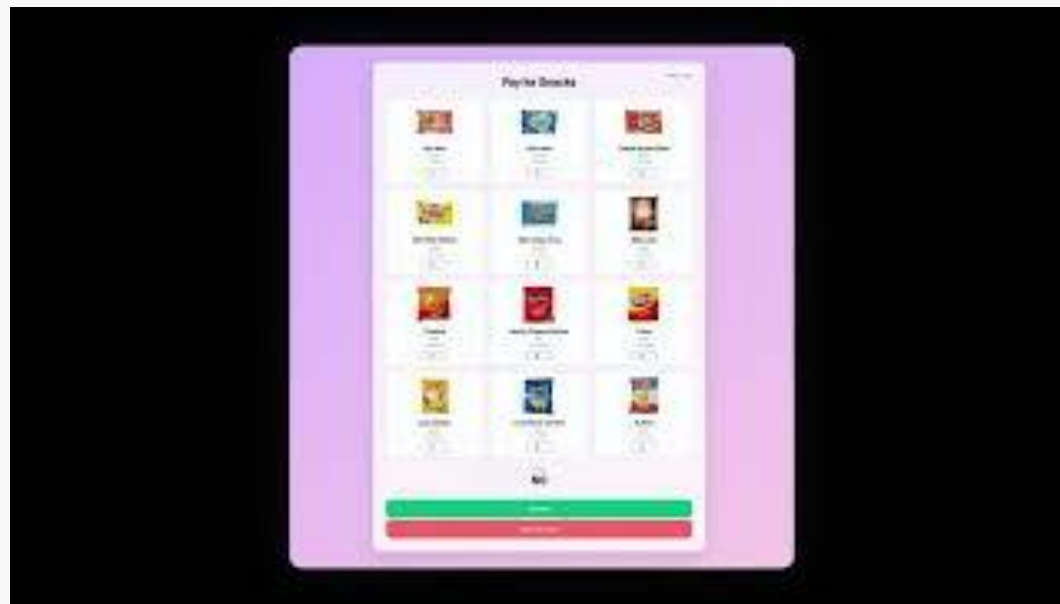
Vending Machine ROBOT

muchin.us





Our Project



WEBSITE TEAM



Website

Frontend Tech Stack

Languages: HTML, Javascript

Cloud: AWS EC2 (running NGINX)

Toolkits: Vite, Tailwindcss



Backend Tech Stack

Language: Python

Backend Framework: Flask + CORS

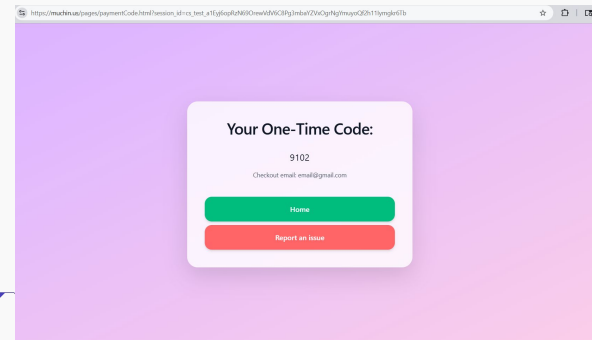
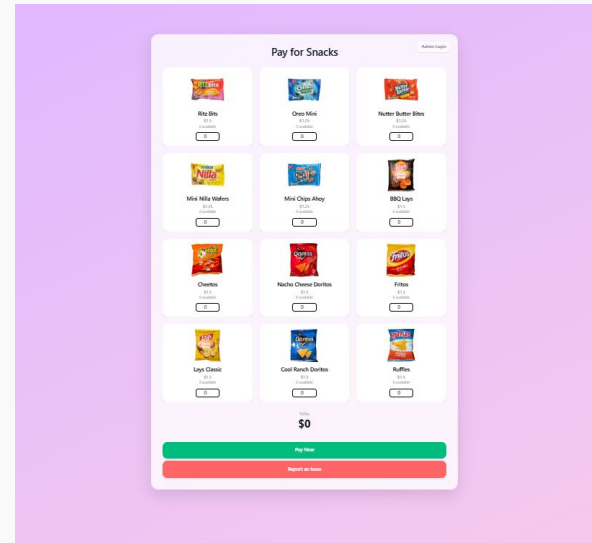
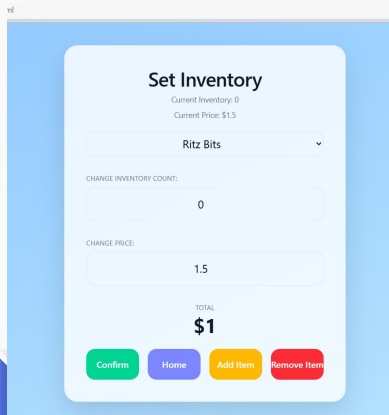
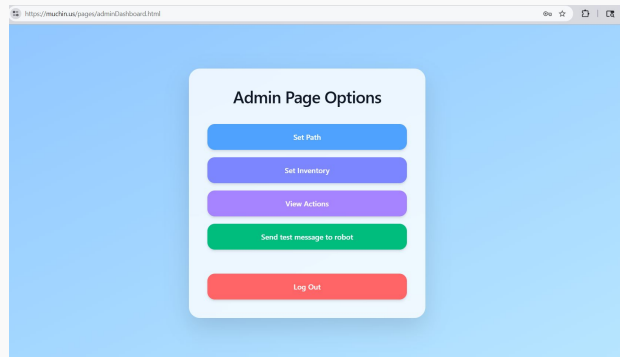
Database: SQLite

Payments: Stripe Checkout + Stripe Webhooks

Security / Auth Utilities: bcrypt

Integrations / Utilities: requests (Discord webhooks), python-dotenv, TCP sockets + threading

Website



Website Challenges

- EC2 Deployment
- Repo/ Code Pipeline
Management



PATHFINDING TEAM



AI/Pathfinding

Hardware

RaspberryPi 4B (*computer*)

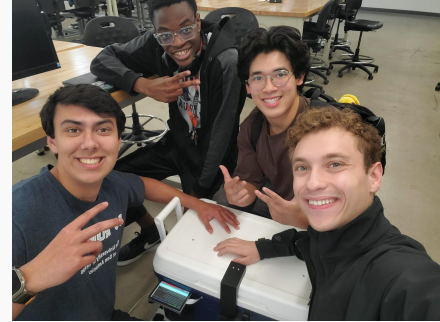
NCS2 (*edge AI chip*)

PiCamera Module 3

GPS

Pololu Maestro (*servo controler*)

Dual H-Bridge Motor Driver Module (*motor controllers*)



AI/Pathfinding

Software

Language: Python

Libraries: threading, queue, time, os, atexit, socket, logging, collections.deque, subprocess, signal, sys, json, math, random, numpy, opencv-python, openvino, pygame, smbus, RPi.GPIO, pigpio, pyserial

Toolkits: OpenVino



Pathfinding Challenges

- Robot did not move
- Axles problem



ENGINEERING TEAM



The Robot

Tools

Software: Solidworks, Fusion

Manufacturing: Welding (MIG), 3D Printing, Soldering

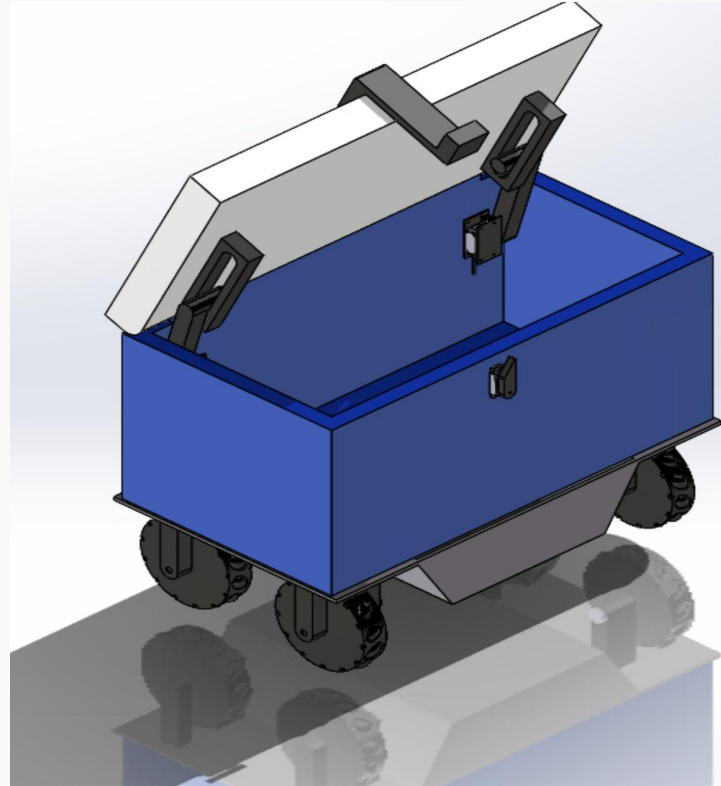
Hardware: M3 and M8 Bolts (used to supplement the strength of PLA with metal)

Materials

24-Gauge Mild Steel

3D-Printed PLA

3D-Printed TPU



Robot Challenges

- Moving robot across campus
- Waiting for parts to arrive



Star our Repos!



<https://github.com/Vendning-Machine-Team>

Thank you!



<https://github.com/Vendning-Machine-Team>

DANKE !

Gibt es noch Fragen?

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ICON PACK



ALTERNATIVE RESSOURCEN

Hier ist eine Auswahl an alternativen Ressourcen im gleichen Stil wie diese Vorlage:

Vektoren

- [Realistic vertical poster template for cyber monday sale](#)

